

Project Title	AI Agentic YouTube Comment Sentiment Analysis and SQL Reporting System
Skills take away From This Project	Learners will gain hands-on experience in Python automation, agent-based workflow design, API integration, SQL data modeling, comment ingestion pipelines, Hugging Face sentiment analysis, data cleaning, aggregation logic, and report generation in any environment.
Domain	Artificial Intelligence, Natural Language Processing, Social Media Analytics, Data Engineering, and Business Intelligence.

Problem Statement:

Build an autonomous Hugging Face-based agent that accepts a YouTube channel as input, identifies the latest 10 uploaded videos, extracts comments for each video, stores structured data in SQL, performs sentiment analysis on every comment, and produces a video-wise and channel-wise sentiment summary that helps users understand audience reactions.

Business Use Cases:

- Marketing teams can measure audience reaction for the latest campaign videos and identify which video receives the most positive or negative public response.
- Content creators can evaluate how viewers react to new uploads and identify repeating praise themes or complaint patterns.

- Brand monitoring teams can track sentiment shifts across a channel over time and detect content that drives engagement risk.
- Media analysts can compare sentiment distributions across different videos to support editorial or production decisions.
- Training institutes can use the system as a practical example of agentic AI, NLP, and SQL integration in one project.

Approach:

1. Accept a YouTube channel handle, URL, or channel ID as input in Colab.
2. Use an agent framework from Hugging Face to orchestrate the full workflow.
3. Resolve the channel, identify the uploads playlist, and fetch the latest 10 uploaded videos.
4. Extract top-level comments for each video using the official YouTube Data API.
5. Store channel, video, and comment data in SQL tables with clear relationships.
6. Apply a Hugging Face sentiment model on each comment and write the results back to SQL.
7. Aggregate the sentiment results into per-video metrics and an overall channel summary.
8. Export final outputs as SQL tables, CSV reports, and a concise analytical summary.

Results:

1. Accept a YouTube channel handle, URL, or channel ID as input in Colab.
2. Use an agent framework from Hugging Face to orchestrate the full workflow.
3. Resolve the channel, identify the uploads playlist, and fetch the latest 10 uploaded videos.

4. Extract top-level comments for each video using the official YouTube Data API.
5. Store channel, video, and comment data in SQL tables with clear relationships.
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7. Aggregate the sentiment results into per-video metrics and an overall channel summary.
8. Export final outputs as SQL tables, CSV reports, and a concise analytical summary.

Metric	What it measures
Pipeline completion rate	Whether the agent successfully completes input resolution, video retrieval, comment extraction, SQL storage, sentiment scoring, and reporting.
Data capture quality	Number of videos fetched, number of comments stored, and percentage of valid non-empty records.

SQL design quality	Correctness of table relationships, primary keys, duplicate handling, and queryability of stored data.
Sentiment coverage	Percentage of collected comments that receive a valid sentiment label and score.
Reporting clarity	How clearly the final output explains positive versus negative sentiment at video level and overall channel level.
Code quality	Readability, modular organization where applicable, reproducibility in Colab, and error handling for unavailable comments or API limits.

Technical Tags:

Python, Hugging Face, smolagents, transformers, YouTube Data API, SQL, SQLite, MySQL, sentiment analysis, NLP, agentic AI, pandas, automation, reporting.

Data Set:

No static dataset is required at the start. The primary dataset is created dynamically by the agent from live YouTube channel metadata, uploaded video details, and user comments obtained through the official YouTube Data API. Data is stored in SQL tables such as channels, videos, and comments.

Data Set Explanation:

The collected dataset contains three major entities: channel-level information, video-level information, and comment-level information. Typical variables include channel_id, channel_title, video_id, video_title, published_at, author_name, comment_text, like_count, sentiment_label, and sentiment_score. Preprocessing steps may include comment text cleaning, null handling, duplicate removal, emoji or symbol normalization, truncation for model input length, and optional language filtering.

Project Deliverables:

- A Colab notebook containing the complete Python implementation.
- SQL database file or SQL dump showing channels, videos, and comments tables.
- CSV exports for videos, comments, and sentiment summary results.
- A short technical document explaining architecture, workflow, tools, and observations.
- A final presentation or demo showing how the agent works from channel input to sentiment report.

Project Guidelines:

- Use Python only and keep the workflow reproducible.
- Use Hugging Face libraries for the agent and sentiment model; avoid paid proprietary AI services.
- Use the official YouTube Data API for video and comment collection.
- Design SQL tables carefully and avoid duplicate inserts by using proper keys or upsert logic.
- Handle edge cases such as comments disabled, API quota limits, missing fields, and empty videos.

- Document configuration steps clearly, including API keys, library installation, and runtime instructions.
- Use Git or another version control method to track notebook and script changes.
- Keep outputs interpretable: do not stop at raw labels; include meaningful percentages and video-wise summaries.

Timeline:

Week	Milestone	Expected outcome
1-4 days	Setup and design	Finalize project scope, create API access, prepare Colab environment, and define SQL schema.
5-8 days	Agent and ingestion	Build the agent workflow to resolve channels, fetch videos, pull comments, and populate SQL tables.
9-12 days	Sentiment and reporting	Run Hugging Face sentiment analysis, create summaries, export outputs, and prepare final documentation/demo.

PROJECT DOUBT CLARIFICATION SESSION (PROJECT AND CLASS DOUBTS)

About Session: The Project Doubt Clarification Session is a helpful resource for resolving questions and concerns about projects and class topics. It provides support in understanding project requirements, addressing code issues, and clarifying class concepts. The session aims to enhance comprehension and provide guidance to overcome challenges effectively.

Note: Book the slot at least before 12:00 Pm on the same day

Timing: Monday-Saturday (3:30PM to 4:30PM)

Session link : [Project Doubt Session \[DS/AIML\]](#)

For BMAI project/class topic doubt slot clarification session:

Session link : [BMAI/BADM_Project Doubt Clarification Session Link](#)

Timing: Tuesday, Thursday and Saturday (3:00PM to 4:00PM)

LIVE EVALUATION SESSION (CAPSTONE AND FINAL PROJECT)

About Session: The Live Evaluation Session for Capstone and Final Projects allows participants to showcase their projects and receive real-time feedback for improvement. It assesses project quality and provides an opportunity for discussion and evaluation.

Note: This form will Open only on Saturday (after 2 PM) and Sunday on Every Week

Timing:

For BMAI and DE

Monday-Saturday (11:30AM to 1:00PM)

For DS and AIML

Timing:

Monday-Saturday (11:30AM to 12:30PM) and (05:30PM to 07:00PM)

Booking link : <https://forms.gle/1m2Gsro41fLtZurRA>

Evaluation Metrics : [Project Live Evaluation](#)

